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Diversity-Oriented Enzymatic Modular Synthesis of *N*-Glycans Bearing LacdiNAc-Related Epitopes

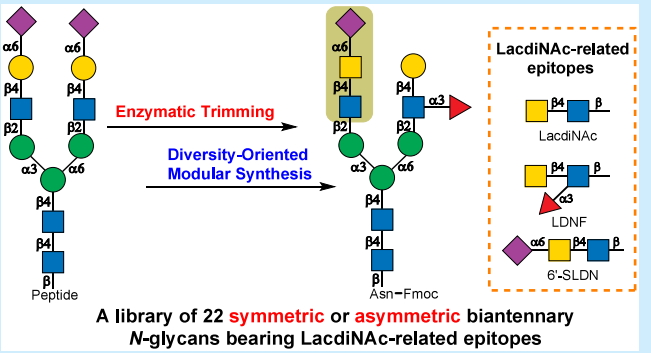
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ABSTRACT: The disaccharide unit GalNAcβ1,4GlcNAc (LacdiNAc or LDN) and its α1,3-fucosylated derivative LDNF and α2,6-sialylated derivative 6′-SLDN are common epitopes of various *N*-glycans. Due to their structural complexity, the synthesis of a comprehensive library of *N*-glycans bearing LacdiNAc-related epitopes remains a significant challenge. Herein, we present a highly efficient enzymatic modular assembly strategy for the collective synthesis of 22 biantennary *N*-glycans containing LacdiNAc, LDNF, or 6′-SLDN epitopes. This strategy features the site-selective enzymatic preparation of two asymmetric *N*-glycan core structures, followed by diversity-oriented derivatization to provide 16 hard-to-obtain asymmetric biantennary *N*-glycans bearing LacdiNAc-related epitopes.

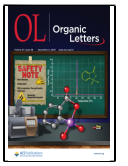


Glycosylation is one of the most pivotal and complex post-translational modifications of proteins, involving diverse physiological processes, such as signal transduction, cell differentiation, neurodevelopment, and immune regulation.[1] Aberrant glycosylation is frequently associated with various pathological conditions, including cancer, autoimmune diseases, inflammatory diseases, and neurodegenerative disorders.[1c,2] The disaccharide unit Galβ1,4GlcNAc (LacNAc) and its derivatives constitute the most prevalent nonreducing terminal modification in glycoproteins, and their biological functions are widely understood.[1c,3] A growing body of evidence reveals that its structural counterpart, GalNAcβ1,4GlcNAc (LacdiNAc or LDN), also serves as a common terminal epitope on *N*-glycans of diverse glycoproteins, including enzymes, hormones, membrane proteins, and transporters.[4] Similar to LacNAc, the LacdiNAc disaccharide unit can undergo further modifications, including fucosylation, sialylation, or sulfation, leading to diverse terminal epitopes such as α1,3-fucosylated LacdiNAc (LDNF), α2,6-sialylated LacdiNAc (6′-SLDN), and 4′-*O*-sulfated LacdiNAc (4′-*O*-sulfo LDN) (Scheme 1a).[5]

LacdiNAc-related *N*-glycans are abundantly expressed in glycoproteins of lower invertebrates,[4a,c] exemplified by the highly fucosylated *N*-glycans bearing LDNF epitopes on the surface of schistosome cells, capable of triggering strong immune responses in humans,[6] making them potential targets for serodiagnosis and antihelminthic vaccine development. In normal mammalian systems, LacdiNAc-modified *N*-glycans exhibit tissue- and protein-restricted expression, being localized to several pituitary hormones, β1-integrin, leukemia inhibitory factor receptor, and tenascin-R.[6a,7] These *N*-glycans play significant roles in the regulation of the levels of circulating hormones and bone mass, the maintenance of cell stemness, and immune homeostasis.[7,8] Moreover, LacdiNAc-type *N*-glycans are widely and differentially expressed across multiple human tumors, which have been recognized as valuable biomarkers for cancer diagnostics.[7b,9] For example, prostate-specific antigen (PSA), which exhibits an increased level of expression of LacdiNAc-containing epitopes on its *N*-glycans, has been widely used as a serum biomarker for the diagnosis of prostate cancer.[10]

Despite the biological and clinical significance of LacdiNAc-type *N*-glycans, only a limited number of LacdiNAc-related *N*-glycans have been synthesized to date (Scheme 1b).[6a,11] One of the previous efforts has successfully delivered symmetric *N*-glycans containing either LacdiNAc or LDNF epitopes (Scheme 1b, **I** and **II**) through enzymatic assembly with recombinant *Caenorhabditis elegans* β1,4-*N*-acetylgalactosaminyltransferase (B4GALNT).[6a] The synthesis of asymmetric *N*-glycans bearing LacdiNAc epitopes faced significant challenges due to the lack of branch selectivity of the known β1,4-*N*-acetylgalactosaminyltransferases and α1,3-fucosyltransferases,

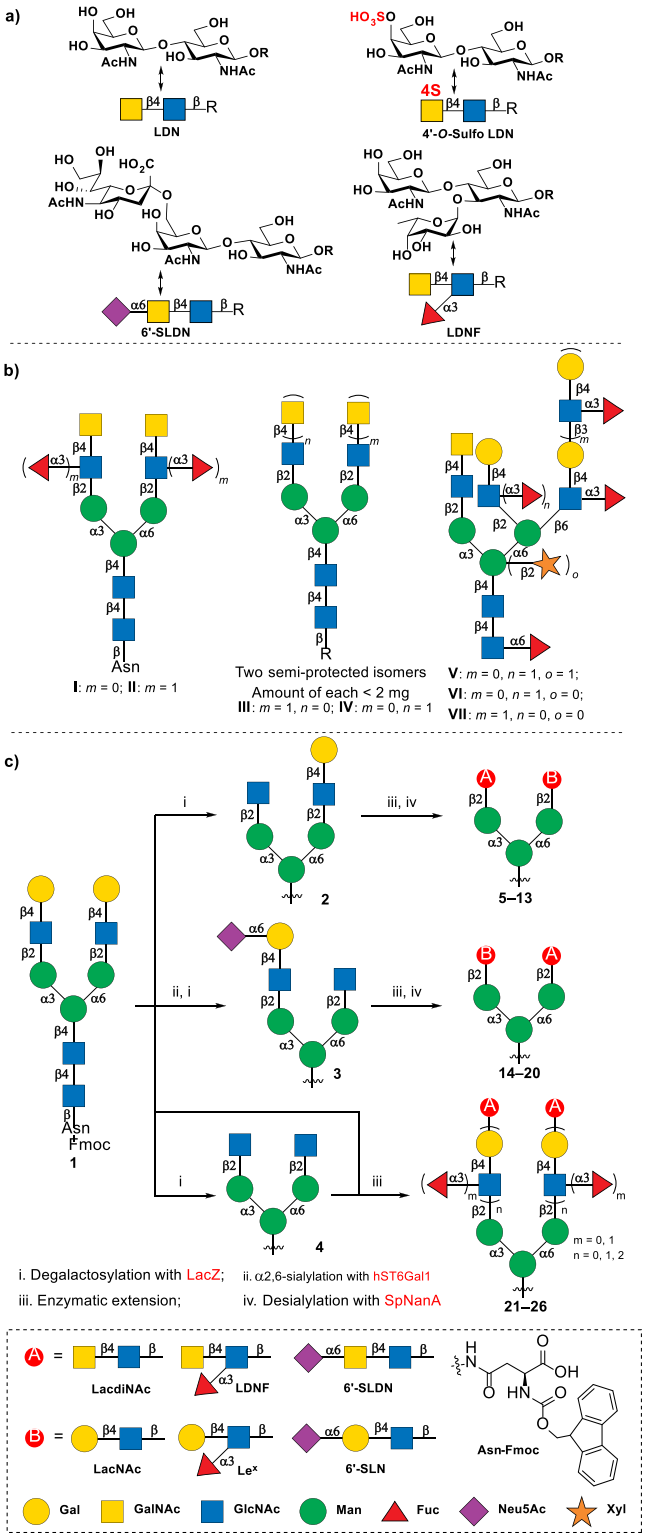
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Scheme 1. (a) Common LacdiNAc-Related Epitopes and (b) Previously and (c) Currently Synthesized LacdiNAc-Related *N*-Glycans



necessitating laborious isolation of positional isomers (Scheme 1b, **III** and **IV**) from reaction mixtures via HPLC, which severely impairs the efficiency and scalability of the preparation of these complex targets.[11a] To avoid this trouble, Boons and co-workers assembled three LDNF-containing asymmetric *N*-glycans (Scheme 1b, **V**–**VII**) from *Schistosoma mansoni*, in which chemical glycosylation was used to install the β1,4-linked GalNAc for LacdiNAc formation.[11b] While their approach provided elegant access to these complex glycans, the chemical synthesis step posed a limitation to the overall efficiency. Hence, a concise and scalable route to LacdiNAc-bearing *N*-glycans is still highly desirable to enable their structural elucidation and functional validation. Notably, asymmetrically branched *N*-glycans that simultaneously carry both LacdiNAc- and LacNAc-related epitopes remain unexplored to date.

Previous studies have demonstrated that *Escherichia coli* galactosidase LacZ preferentially cleaves the terminal Gal from the α1,3Man branch of symmetric biantennary *N*-glycan **1**,[12] while human α2,6-sialyltransferase ST6Gal1 (hST6Gal1) selectively installs a sialic acid onto the identical α1,3Man branch Gal residue.[13] As shown in Scheme 1c, leveraging the selectivity of these enzymatic transformations, we herein develop an efficient enzymatic assembly strategy that starts from readily available symmetric *N*-glycan **1** and generates key asymmetric intermediates (**2** or **3**) through site-specific degalactosylation or sialylation. LacdiNAc-related epitopes (A being LacdiNAc, LDNF, or 6′-SLDN) were then introduced onto the α1,3Man branch of **2** and the α1,6Man branch of **3**, and the remaining antenna was extended with either LacNAc-related epitopes (B being LacNAc, Le^x, or 6′-SLN) or additional LacdiNAcs, thereby enabling the diversified synthesis of asymmetric biantennary *N*-glycans **5**–**20**. Additionally, six symmetric biantennary *N*-glycans **21**–**26** were synthesized to further expand this *N*-glycan library. The present synthetic approach enabled the first synthesis of 18 LacdiNAc-type *N*-glycans, including **5**–**18** and **23**–**26**.

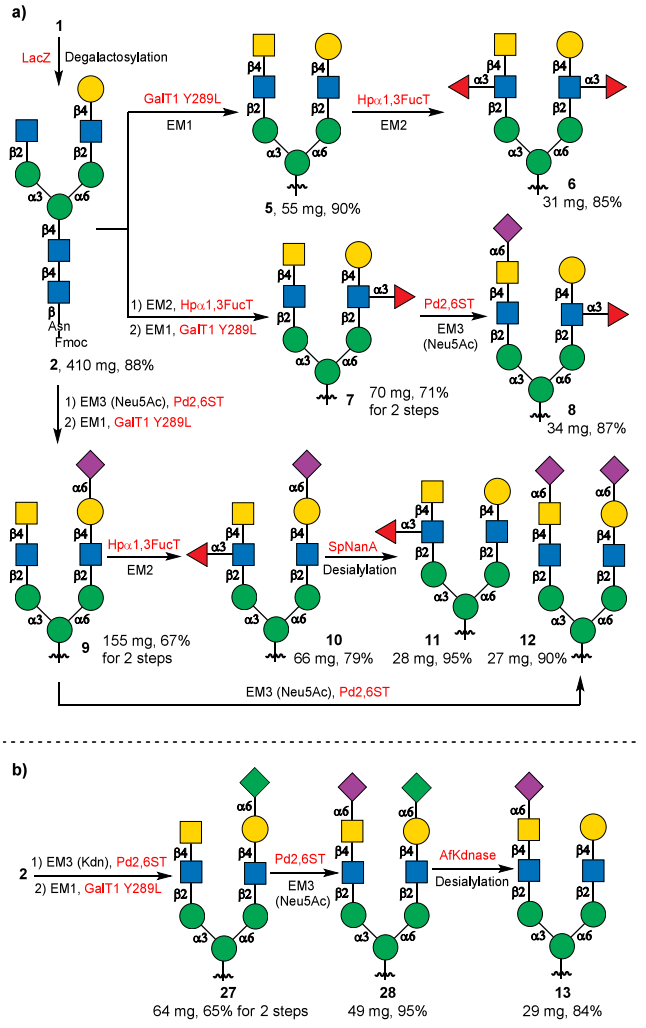
The complex structures of *N*-glycans pose fundamental challenges for their *de novo* synthesis.[14] Sialoglycopeptides (SGP), a type of glycoconjugate containing complex-type *N*-glycans, can be isolated from readily available egg yolk powder and have been widely utilized as starting materials for the semisynthesis of *N*-glycans.[15] Following a reported method,[16] the SGP mixtures were isolated and then converted into homogeneous biantennary *N*-glycan **1** on gram scales through the sequential steps of peptide cleavage, Fmoc-amino protection, desialylation, and galactosylation (Scheme S10).[17]

Asymmetric biantennary *N*-glycans **5**–**13**, bearing LacdiNAc, LDNF, or 6′-SLDN epitopes at the α1,3Man branch, were prepared as shown in Scheme 2. The synthesis commenced with the selective removal of the Gal residue at the α1,3Man branch of nonasaccharide **1** using LacZ, affording key intermediate octasaccharide **2** in 88% yield. The high selectivity of this enzymatic degalactosylation is ensured by deliberate control of the reaction conditions, including enzyme concentration (<0.6 mg/mL), substrate concentration (<7 mg/mL), and reaction time (<8 h). Several β1,4-*N*-acetylgalactosaminyltransferases capable of synthesizing the LacdiNAc disaccharide structure have been functionally characterized, including *C. elegans* B4GALNTs[18] and a bovine β1,4-galactosyltransferase mutant (GalT1 Y289L).[19] Among these, the bovine GalT1 Y289L mutant, demonstrating high-level expression in *E. coli* systems with excellent stability and catalytic efficiency, was employed for LacdiNAc unit construction in this study. In enzyme module 1 (EM1), a β1,4-linked GalNAc was introduced onto the α1,3Man branch GlcNAc terminus of **2** with GalT1 Y289L in MES buffer (50 mM, pH 7.5) for 8–12 h, resulting in the formation of nonasaccharide **5** containing a LacdiNAc group in 90% yield.

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Scheme 2. Synthesis of Asymmetric Biantennary *N*-Glycans 5−13



The subsequent α1,3-fucosylation of **5** with EM2[20] yielded undecasaccharide **6**, which harbors both LDNF and Le^x epitopes, in 85% yield. Alternatively, octasaccharide **2** was first α1,3-fucosylated using EM2, followed by further β1,4-*N*-acetylgalactosaminylation with EM1, providing decasaccharide **7** in more than 70% yield over two steps. The α1,3Man branch GalNAc terminus of **7** was then α2,6-sialylated with EM3[21] in the presence of Neu5Ac as a donor precursor, affording undecasaccharide **8** bearing the 6′-SLDN epitope in 87% yield.

Glycans **9**−**12** were also derived from octasaccharide **2** (Scheme 2a). Initially, the α1,6Man branch Gal terminus of **2** was modified with Neu5Ac by using EM3. The resulting intermediate was then β1,4-*N*-acetylgalactosaminylated with EM1, affording decasaccharide **9** in 67% yield over two steps. Notably, the introduction of α2,6-linked Neu5Ac at the α1,6Man branch Gal residue sterically hinders α1,3-fucosylation at the adjacent LacNAc unit,[22] restricting the α1,3-fucosylation exclusively to the α1,3Man branch GlcNAc residue of **9**, providing undecasaccharide **10** in 79% yield. The subsequent treatment of **10** with sialidase SpNanA cleanly removed the Neu5Ac residue to afford decasaccharide **11** in 95% yield. Meanwhile, the LacdiNAc group of **9** was further converted into a 6′-SLDN epitope with EM3, leading to the formation of disialylated undecasaccharide **12** in 90% yield.

Although hST6Gal1 selectively introduces α2,6-linked sialic acid onto the α1,3Man branch, it cannot utilize GalNAc glycosides as substrates. Pd2,6ST is a bacterial α2,6-sialyltransferase with promiscuous substrate specificity, which can tolerate both Gal and GalNAc glycosides for sialylation.[23] However, the promiscuous substrate specificity of Pd2,6ST blocked direct access from **5** to **13** via α2,6-sialylation with EM3. We recently developed a ketodeoxynonulosonic acid hydrolase (Kdnase)-assisted enzymatic substrate engineering strategy for site-selective α2,6-sialylation of complex substrates containing multiple Gal/GalNAc residues using Pd2,6ST.[24] Applying this approach, decasaccharide **13** was synthesized from **2** in a site-selective manner. As shown in Scheme 2b, α2,6-linked ketodeoxynonulosonic acid (Kdn) was first introduced onto the Gal residue of **2** as a temporary protecting group. The resulting intermediate was then elongated with β1,4-linked GalNAc using EM1, affording decasaccharide **27** in 65% yield over two steps. The subsequent α2,6-sialylation of **27** with Neu5Ac yielded undecasaccharide **28**, followed by the orthogonal removal of the Kdn protecting group by Kdnase from *Aspergillus fumigatus* (AfKdnase),[24,25] affording decasaccharide **13** in 80% yield over two steps.

Scheme 3 outlines the synthesis of *N*-glycans **14**−**20** containing LacdiNAc-related epitopes at the α1,6Man branch. The key sialyltransferase in this synthetic route, hST6Gal1, has been heterologously expressed in prokaryotic systems such as *E. coli*,[26] thereby significantly facilitating its application in synthesis. Thus, the synthesis commenced with selective α2,6-sialylation of the α1,3Man branch Gal terminus in nonasaccharide **1** using hST6Gal1. The subsequent degalactosylation provided key intermediate **3** in 74% yield over two steps, which was converted into decasaccharide **14** through β1,4-*N*-acetylgalactosaminylation to form a LacdiNAc group at the α1,6Man branch in 84% yield (Scheme 3a).

Further derivatization of decasaccharide **14** provided remaining *N*-glycans **15**−**20** (Scheme 3b). The removal of the Neu5Ac residue from **14** provided nonasaccharide **15**, which was then subjected to α1,3-fucosylation with EM2, giving difucosylated decasaccharide **16** in 82% yield over two steps. Alternatively, decasaccharide **14** was α1,3-fucosylated with EM2 to give undecasaccharide **17**, which contains the LDNF epitope, in 95% yield. Obtained compound **17** underwent further desialylation with SpNanA to furnish decasaccharide **18** in 97% yield. The structural reconfiguration of **18** using LacZ and EM1 afforded decasaccharide **20** bearing both LacdiNAc and LDNF epitopes in 83% yield over two steps. By implementation of a rational enzymatic assembly strategy, all asymmetric biantennary LacdiNAc-type *N*-glycans **5**−**20** were synthesized site-selectively, obviating the need for HPLC purification to separate isomeric mixtures.

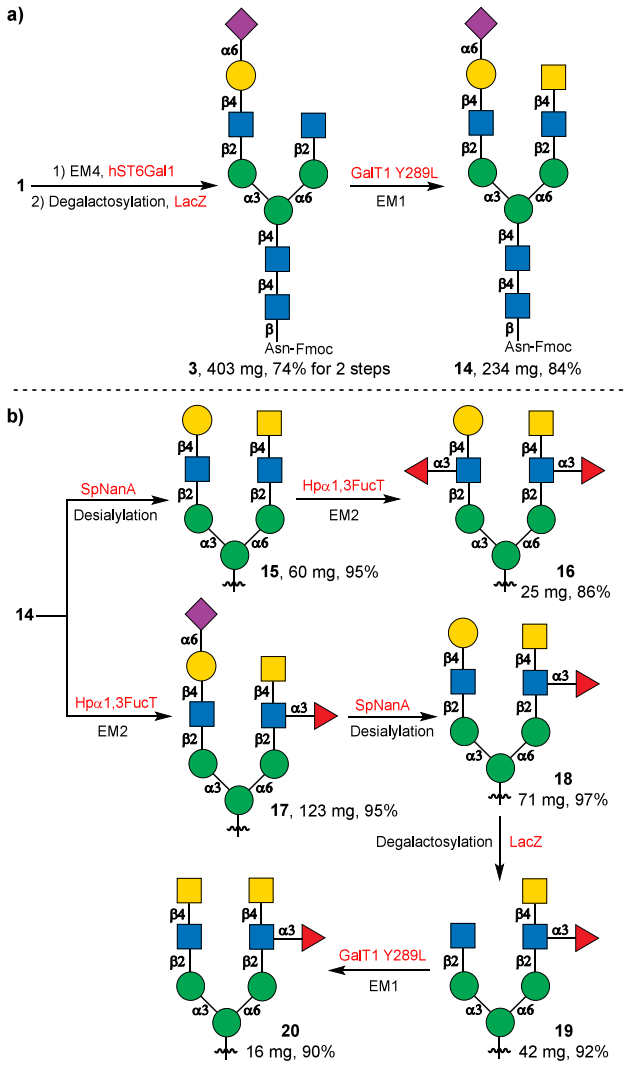
The synthesis of symmetric biantennary *N*-glycans **21**−**23** starting from nonasaccharide **1** is illustrated in Scheme 4. Both LacNAc units of **1** were converted into LacdiNAc units via enzymatic trimming with LacZ and GalNAc supplementation with EM1, affording nonasaccharide **21** in 76% yield over two steps. Obtained **21** underwent further α1,3-fucosylation or α2,6-sialylation, leading to the formation of undecasaccharide **22** or **23**, each having two LDNF or two 6′-SLDN epitopes, respectively.

Poly-LacNAc motifs are ubiquitous in natural glycans and glycoconjugates,[22] whereas chimeric LacdiNAc−LacNAc motifs have not yet been reported in natural *N*-glycans to the best of our knowledge. Since the underlying glycan

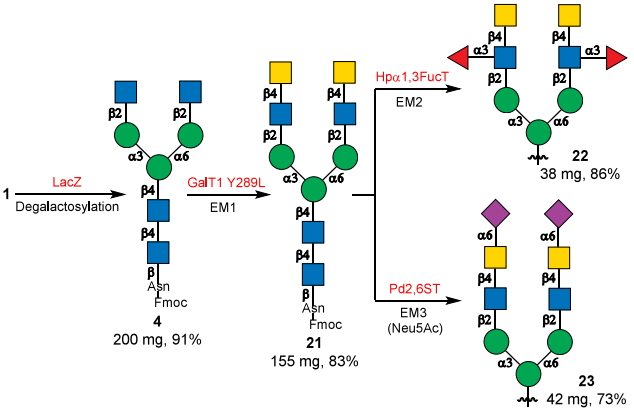
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Scheme 3. Synthesis of Asymmetric Biantennary *N*-Glycans 14−20

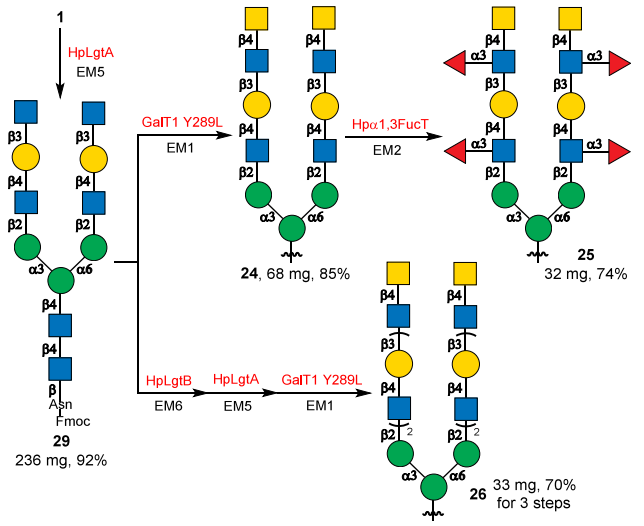


Scheme 4. Synthesis of Symmetric Biantennary *N*-Glycans 21−23



structure could influence the interaction between terminal epitopes and glycan-binding proteins, we aimed to synthesize complex-type *N*-glycans **24**−**26** that contain LacdiNAc−LacNAc-related motifs. As shown in Scheme 5, nonasaccharide **1** was first β1,4-*N*-acetylglucosaminylated using EM5 to afford

Scheme 5. Synthesis of Symmetric Biantennary *N*-Glycans 24−26



intermediate **29** in 92% yield. This intermediate was then sequentially modified through β1,4-*N*-acetyl-galactosaminylation and α1,3-fucosylation to give tridecasaccharide **24** and its tetrafucosylated derivative **25** in excellent yields. Alternatively, both antennas of **29** were simultaneously elongated sequentially with EM6,[20] EM5, and EM1, affording heptadecasaccharide **26** bearing two LacdiNAc−$(LacNAc)_2$ termini in 70% yield over three steps.

In summary, we have developed a highly efficient enzymatic strategy for the diversity-oriented synthesis of 22 biantennary *N*-glycans containing LacdiNAc, LDNF, or 6′-SLDN epitopes. Leveraging the unique substrate specificities of LacZ and hST6Gal1 and the elaborately designed enzymatic assembly routes, LacdiNAc-related epitopes could be flexibly installed at the α1,3Man or α1,6Man branch of the *N*-glycan core structures, thereby enabling the collective and site-selective synthesis of 16 asymmetric biantennary *N*-glycans. These homogeneous synthetic glycans provide well-defined standards for the structural characterization of LacdiNAc-type *N*-glycans, as well as the investigation of their structure−activity relationships with glycan-binding proteins such as lectins and antibodies.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c04338.

Experimental procedures, characterization data, and supplementary schemes, figures, and HRMS and NMR spectra (PDF)

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Notes

The authors declare no competing financial interest.

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